

PROMPT FOR RESOLVING FRANKL'S UNION-CLOSED SETS CONJECTURE

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ABSTRACT. This document contains a full frontier-model research prompt for resolving the union-closed sets conjecture. The target is the exact one-half frequency threshold for every finite nontrivial union-closed family, or a fully verified finite counterexample. Status snapshot: 20 July 2026.

1. Prompt

Current task statement

A finite family of sets $\mathcal{F}$ is union-closed if $A \cup B \in \mathcal{F}$ whenever $A, B \in \mathcal{F}$. Frankl's conjecture states that every finite union-closed family containing at least one nonempty set has an element that belongs to at least half of its member sets:

$$\exists x \in \bigcup \mathcal{F} : |\{A \in \mathcal{F} : x \in A\}| \geq \frac{|\mathcal{F}|}{2}.$$

Resolve this statement completely. The empty set may belong to $\mathcal{F}$; duplicate sets do not. The ground set is understood to be $\bigcup \mathcal{F}$, so unused elements are irrelevant.

Assume for purposes of this task that a complete resolution is reachable. Do not assume whether the conjecture is true or false.

Complete-resolution standard

A complete resolution must establish one alternative:

- Affirmative alternative: prove the frequency-1/2 conclusion for every finite union-closed family with a nonempty member;
- Negative alternative: provide an explicit finite union-closed family in which every element occurs in strictly fewer than half of the sets, and prove all conditions exactly;
- in either direction, justify all reductions such as separating families, deletion of unused elements, compression, lattice representation, or minimality.

What does not count

Partial progress is insufficient unless it logically implies the exact resolution above. In particular, the following do not count:

- improving the best universal constant below 1/2 without reaching 1/2;
- proving the conjecture only for bounded ground-set size, bounded family size, regular families, separating families without a proved reduction, or other special classes;
- computational verification through any fixed size;
- an entropy, averaging, or linear-programming inequality with an unproved extremal assumption;
- a multiset family, approximately union-closed family, or family whose claimed closure omits some pair;
- a reduction to an equivalent or stronger conjecture with no new proof of that conjecture.

Use multiagent v2 aggressively and dynamically. You have up to 64 concurrent agents available. Do not use a fixed assignment such as "N agents for strategy X." Instead, manage the search using the following heuristics:

- Begin with a genuinely diverse portfolio of approaches. Agents should explore substantially different formulations, invariants, reductions, algebraic viewpoints, structural inductions, decompositions, optimization methods, exact computational searches, and formal-proof routes.
- Do not tell most agents the currently favored approach. Preserve independence during early rounds so that agents do not all converge to the same attractive but incomplete reduction.
- Maintain an explicit registry of approach families. Group agents by the mathematical mechanism they are using, not by superficial wording. If too many agents converge to one family, redirect some toward underexplored formulations.
- Do not allow one route to dominate merely because it gives elegant reductions. A route that ends at a missing lemma equivalent in strength to the original problem is not close to completion unless it supplies a genuinely new proof of that lemma.
- When an approach stalls at a theorem-strength missing lemma, mark that route as blocked. Reopen it only when an agent proposes a materially new mechanism, invariant, construction, or exact experiment.
- Keep several incompatible proof or construction routes alive through multiple rounds. Cross-pollinate ideas only after independent agents have developed them far enough to expose their real strengths and gaps.
- Use adversarial agents throughout. Every candidate theorem, construction, certificate, reduction, and program must be attacked independently. Assign separate agents to mathematical correctness, completeness of case analysis, numerical exactness, implementation soundness, and novelty checking.
- Require agents to return concrete lemmas, constructions, equations, counterexamples to proposed sublemmas, executable encodings, or formally stated proof obligations. Reject status reports, vague optimism, and claims that a global compatibility step is "routine."
- The root agent should repeatedly synthesize, challenge, redirect, and launch new rounds. Do not stop after the first wave fails. Convert promising patterns into exact conjectures and send those conjectures to independent proof and falsification teams.

Problem-specific search portfolio

In the initial independent rounds, ensure that the portfolio includes, but is not limited to, the following genuinely different families:

- minimal-counterexample structure under element deletion, set deletion, quotienting identical incidence columns, and lattice or semilattice decompositions;
- entropy methods, correlated random sets, OR-couplings, information inequalities, and sharpened versions of existing constant-factor arguments;
- compression, shifting, down-set/up-set duality, averaging over maximal chains, and Boolean-lattice isoperimetry;
- weight functions, separating reductions, local configurations, abundant elements, and induction using a quantitative potential with genuine slack;
- linear and semidefinite programming over incidence patterns, flag-algebra-style local statistics, and exact dual certificates that suggest universal inequalities;
- integer programming and SAT searches for minimal counterexamples, used to infer forbidden configurations or finite bases rather than only extending a size bound;
- connections with closure systems, finite lattices, hypergraphs, bipartite incidence graphs, submodularity, and additive combinatorics;
- constructive counterexample search using algebraic, product, blow-up, substitution, and semilattice operations, with exact frequency tracking.

Mandatory adversarial audit

Any surviving candidate must be checked against every item below by agents that did not originate the candidate:

- check union closure for every ordered or unordered pair of member sets, including a set paired with itself;
- check the family is a set of distinct sets and contains at least one nonempty member;
- compute frequencies relative to exactly $|\mathcal{F}|$ and treat equality at one-half correctly;
- audit any reduction to separating families or normalized ground sets in both directions;
- ensure an induction decreases a well-founded parameter and does not lose the target element when

reconstructing the family;

- convert numerical optimization outputs to exact rational inequalities and prove all local constraints are globally realizable or universally valid;
- for a proposed counterexample, use two independent representations and verifiers to rule out omitted unions, duplicates, and frequency errors.

Required final artifact

The final response must provide the full mathematical argument and exact supporting data:

- for a proof, a complete theorem chain from arbitrary finite union-closed families to the $1/2$ conclusion;
- for a counterexample, a canonical machine-readable list of all sets, plus a verifier for closure and frequencies;
- all computational certificates, LP/SDP duals, SAT proofs, or finite classifications used in the argument;
- source code, hashes, exact commands, and an independent audit report;
- a novelty check against the established union-closed literature and known equivalent formulations.

Public search may be used for standard union-closed theory, known lower constants, and named reductions. Do not search for a purported final proof or counterexample and do not answer by merely restating the best current constant.

Do not return merely because current approaches fail or agents report theorem-strength gaps. Continue launching new rounds, reopening blocked approaches only when there is a genuinely new mechanism, and searching for fresh formulations.

Return only when a complete resolution has been found and survives adversarial audit. Do not return a reduction, a special case, an isolated missing lemma, an improved bound that leaves the target unresolved, a "best effort" literature review, or an explanation of why the problem is difficult.

Spend at least 8 hours of sustained search before even considering return or termination. If the execution environment imposes a hard stop before a complete resolution, do not fabricate success or elapsed work: preserve all machine-checkable artifacts and report only the strongest rigorously verified derivation together with the exact remaining gap.